

STANDARD OPERATING PROCEDURE

Boiler B-202 — Startup, Shutdown and Safety

SOP No: SOP-UTL-012 | Revision: 3 | Date: 01/01/2024

Approved by: Safety Officer Anil Mehta | Valid Until: 31/12/2024

1. Scope

This SOP covers safe startup and shutdown procedures for Boiler B-202 located in the Utilities section. All operators must follow these steps strictly. Non-compliance may result in explosion risk or steam leak.

2. Safety Precautions

Ensure all pressure relief valves PRV-101 and PRV-102 are functional before startup. Verify water level in drum — minimum 50% level required. Check fuel gas pressure at FCV-301 — must be between 0.8 to 1.2 bar. Wear heat-resistant PPE. Permit to Work (PTW) must be obtained before any hot work near Boiler B-202.

3. Startup Procedure

Step 1: Open feedwater inlet valve FWV-01 — verify drum level reaches 60%.

Step 2: Start feedwater pump FWP-10 — check outlet pressure 5.5 bar.

Step 3: Open fuel gas valve FCV-301 slowly — confirm pressure 1.0 bar.

Step 4: Ignite burner — verify flame presence within 5 seconds.

Step 5: Monitor drum pressure — raise to operating pressure of 8.5 bar over 30 minutes.

Step 6: Open main steam valve MSV-01 after pressure stabilizes.

4. Compliance References

This SOP complies with Factory Act 1948 Section 31, OISD Standard 116, and IBR Regulations 1950. Annual inspection by IBR inspector is mandatory. Last IBR inspection: 15/03/2024. Next IBR inspection due: 15/03/2025.